Unified Anomaly Detection via Multi-Scale Contrasted Memory

Loïc Jézéquel, Jean Beaudet, Aymeric Histace, and Ngoc-Son Vu

***Abstract*—Deep anomaly detection aims to provide robust and efficient classifiers for zero-shot (unsupervised, UNS) and few-shot (imbalanced supervised, IMS) settings. However, current models still struggle on edge-case normal samples and are often unable to keep high performance over different scales of anomalies. Additionally, there is a lack of a unified framework that efficiently addresses both UNS and IMS settings. To address these limitations, we present a novel two-stage method which leverages multi-scale normal prototypes during training to compute an anomaly deviation score. First, we employ a novel memory-augmented contrastive learning to jointly learn representations and memory modules across multiple scales. This allows us to effectively capture subtle features of normal data while adapting to varying levels of anomaly complexity. Then, we train an efficient anomaly distance-based detector that computes spatial deviation maps between the learned prototypes and incoming observations. Our model outperforms the SoTA on a wide range of anomalies, including object, style, and local anomalies, as well as industrial inspection and face anti-spoofing, while being on par with SoTa out-of-distribution detectors. Notably, it stands as the first model capable of maintaining exceptional performance across both settings.**

***Index Terms*—Anomaly detection, industrial inspection, face anti-spoofing, out-of-distribution, unified model, contrastive learning, differentiable memory, Hopfield network, unsupervised, imbalanced supervised settings.**

I. Introduction

DETECTING deviations from a normal baseline is a pivotal challenge in machine learning and computer vision. However, collecting and annotating anomalous data can be prohibitively expensive. Thus, unsupervised anomaly detection (UNS-AD) has gained significant attention [1], [2], [3], [4]. UNS-AD methods train models exclusively with anomaly-free images to detect unseen anomalies, also termed one-class or zero-shot AD. Yet, in many real-world scenarios, a small set of anomaly samples may become available over time, prompting the need for their inclusion to improve AD performance. Consequently, we transition from UNS to imbalanced supervised (IMS) or few-shot settings. In this context, available anomaly samples are often intricate, incomplete, and ill-defined, leading to a significant departure of AD from conventional binary classification.

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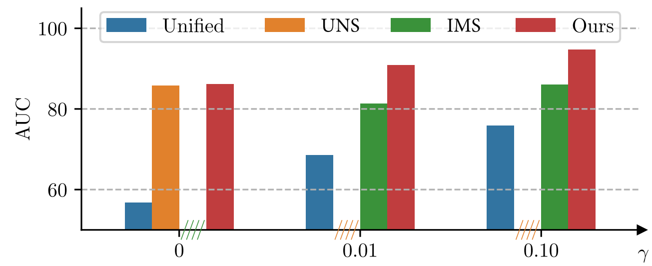

Fig. 1. Comparison of AnoMem with SoTA methods on CIFAR-100 with different anomaly ratio γ. AnoMem is the first unified model to perform well in both UNS and IMS settings. The so-called unified method here exhibits rather random performance for the UNS setting.

To date, no efficient unified AD framework has been able to seamlessly address both UNS and IMS detection. Existing methods typically perform only within their original design settings, as can be seen in Fig. 1. Indeed, these methods employ different specialized approaches with distinct sets of hyperparameters. The addition of anomalies into the training leads to a substantial overhaul of the AD algorithms. This impracticality becomes apparent in real-world applications, especially when only normal samples are initially available (UNS), and the acquisition of anomaly samples occurs at a later stage (IMS).

Apart from the lack of a unified method, existing AD models suffer from additional limitations. First, there exists a hard trade-off between remembering edge-case normal samples and maintaining generalizability toward anomalies, leading to a lack of normal sample memorization and high false reject rates on harder samples. Second, these models often focus on either local low-scale anomalies or global object-oriented anomalies but fail to combine both (see Table I). Furthermore, current models tend to be highly dataset-dependent and do not explicitly utilize multi-scaling.

To overcome these limitations, we present a novel two-stage AD model, dubbed AnoMem. Our approach improves the AD landscape by incorporating multi-scale normal class prototypes within its training phase to compute anomaly deviation scores across various scales. Unlike previous memory bank equipped methods [5], [6], AnoMem leverages normal memory layers spanning multiple scales, thereby augmenting both anomaly detection efficacy and the quality of learned representations. Moreover, by employing modern and ***backpropagable*** Hopfield layers for memorization, our method surpasses the efficiency of nearest neighbor anomaly detectors [7], [8]. While the mentioned detectors require retaining the entire normal set, our model adeptly learns the most representative

TABLE I

OVERVIEW OF ANOMEM'S PERFORMANCE. ANOMEM DEMONSTRATES EXCEPTIONAL CAPABILITIES FOR IMAGE-LEVEL AD WITH DIFFERENT TYPES OF ANOMALY. CATEGORIZATION IS APPROXIMATE: "CLASSIC" AD ARE OBJECT-ANOMALY DETECTORS BETWEEN 2019 AND 2021 (EXCEPT MSC [9] IN 2023), "IAD" DENOTES MORE RECENT METHODS ORIENTED TOWARDS INDUSTRIAL INSPECTION. PERFORMANCE LEVELS ARE QUANTIZED AND DENOTED AS FOLLOWS: ★★★: EXCELLENT (SOTA), ★★: GOOD (COMPARABLE TO SOTA), ★: POOR TO AVERAGE, GRAYED-OUT: CANNOT BE APPLIED, -: RESULTS NOT AVAILABLE OR CANNOT BE APPLIED. RESULTS IN BLUE WERE RE-EVALUATED. "EK" INDICATES METHODS WHICH GET HELP FROM EXTERNAL KNOWLEDGE, SUCH AS PRE-TRAINED MODELS OR EXTRA DATASETS. ECENT VLM-BASED AD METHODS ARE MULTIMODAL AND EK-DEPENDENT, TYPICALLY EVALUATED ON INDUSTRIAL BENCHMARKS. ALTHOUGH LABELED AS ZERO-SHOT, THESE METHODS MAY IMPLICITLY ENCODE ANOMALY KNOWLEDGE THROUGH THEIR PRETRAINED MODALITIES (SEE SECTION II-B)

Setting	Classic AD		IAD-oriented	OOD-oriented		Unified
EK	✓		✓	✓		
Methods	MSC [9]	MHROT [10], GOAD [11]	PatchCore [2], DRAEM [12], DeSTS [13], [14]	CSI [15]	[16]	AnoMem
Object AD	★★★	★★	-	★★★		★★★
Fine-grained AD	-	★	-	★		★★★
Industrial defect	★	-	★★★	-		★★★
FAS (face anti-spoofing)	★★	★	-	★		★★★
OOD		★		★★	★★★	★★

samples within a ***fixed*** size. Through extensive experiments, we demonstrate that our method significantly outperforms all previous memory-equipped anomaly detectors.

Our main contributions are the following:

- We propose to integrate backpropagable memory modules into contrastive learning (CL) to learn normal class prototypes during training. In the first stage, we *jointly* learn representations and memory modules using CL, allowing for effective ***memorization of normal prototypes.***[1] In the second stage, we learn to detect anomalies using the learned prototypes. When anomalies are available, we train a detector on the spatial deviation maps between learned prototypes and observations. AnoMem is the first working well in both UNS and IMS settings with a few anomalies (**unified method**).
- AnoMem is further improved by using multi-scale normal prototypes for representation learning and AD. We introduce a novel way to efficiently memorize 2D features maps spatially. This enables our model to accurately detect low-scale, texture-oriented anomalies and higher-scale, object-oriented anomalies (**generic detector**).
- The efficiency of AnoMem, was validated through extensive experiments with no external knowledge, including different types of anomalies (object, subtle style, local, fine-grained), two real-world applications (industrial inspection and face anti-spoofing), and OOD detection. We improve detection with up to 50% error relative improvement on object anomalies and 14% on face anti-spoofing.

II. RELATED WORK

A. Memory Modules

A memory module should achieve two main operations: ① writing inside a memory from a given set of samples, and ② retrieving the most similar sample from its memory with minimal error when presented with a partial or corrupted input. Memory modules will vary based on their capacity to memorize data given a model size and the average reconstruction error.

A simple memory module is the nearest neighbor queue (***non-differentiable*** memory). Given a maximum size M, it stores the last M samples representations in the queue. To remember an incomplete input x, it retrieves the nearest neighbor from the queue.

A more powerful and expressive form of memory, known as the ***modern differentiable Hopfield layer*** [17], significantly extends classical Hopfield networks. It represents memory as a learnable weight matrix $\boldsymbol{X} \in \mathbb{R}^{d \times N_{\text{Mem}}}$, where each column ξ_i denotes a prototype pattern in a d-dimensional space. Retrieval is performed via the iterative update:

$$\xi^{(t+1)} = \text{softmax}\left(\beta\, \xi^{(t)} \boldsymbol{X}\right) \boldsymbol{X}^{\top}, \tag{1}$$

where $\xi^{(0)}$ is the input query vector input, and β is the inverse temperature that controls the sharpness of the attention. Iteration continues until convergence, i.e., $|\xi^{(t+1)} - \xi^{(t)}| < \epsilon$, where the variable t denotes the iteration step. This update is formally equivalent to the self-attention mechanism in Transformers, but is reapplied recursively, enabling convergence to fixed-point attractors. This allows the Hopfield layer to function as both an associative memory and an attention mechanism.

Key advantages of the Hopfield layer include its ***exponentially large memory capacity***, ***efficient convergence*** in a few steps, and ***low redundancy*** in learned patterns due to softmax-based retrieval. It supports diverse attractors (e.g., global, local, or specific patterns) and integrates seamlessly into modern deep architectures. Henceforth, we refer to this module as a *Hopfield layer of size* N_{Mem}, referring to the number of memory vectors it contains.

B. Anomaly Detection

1) Overview of AD: Due to its importance, AD has received significant attention, leading to a wide variety of proposed methods. For comprehensive surveys of deep AD techniques,

[1] Unlike coreset algorithm, which stores samples directly, we leverage the modern Hopfield network to learn and represent prototypes of normal samples.

we refer readers to [18], [19], while this section focuses on approaches most relevant to our work.

Recent advances have increasingly explored the use of vision-language models (e.g., CLIP [20]) for zero-shot AD. These models inherently rely on EK, such as pre-trained multimodal embeddings, and will be discussed in more detail later. Such developments complement the three major families of AD techniques—pre-trained feature adaptation, discriminative, and generative methods—as categorized in [18]. In addition to methodological categories, AD approaches can be further classified by their supervision levels, depending on the availability and granularity of anomaly annotations. Common settings include unsupervised (UNS), imbalanced supervised (IMS), semi-supervised, and open-set detection. For example, semi-supervised AD leverages partially labeled data, while open-set supervised AD [21] targets previously unseen anomalies despite limited anomaly types being available during training. Recently, the UniAD setting [22] was introduced, enabling a single model to handle multi-class normal samples, avoiding the need to train separate models per object category. While these multi-class UNS-AD methods aim for category generalization, our approach performs robustly in both UNS and IMS settings with few anomaly samples, and effectively detects both texture- and object-level anomalies across different scales.

Concerning its applications and evaluation protocols, AD can also be categorized differently, as depicted in Table I. Early AD methods were evaluated on object-anomaly datasets such as CIFAR10/100, which we refer to as classic AD. The introduction of the MVTec-AD dataset has catalyzed the development of methods for industrial inspection, and recent AD methods are evaluated on industrial datasets rather than CIFAR10/100. Very closely related to AD is out-of-distribution (OOD) detection (we will briefly discuss the difference later). For the OOD detection task, many methods have been recently proposed, but most of them cannot be applied to object-anomaly or industrial inspection. *This paper aims to develop a* ***generic*** *AD, and we evaluate AnoMem on these evaluation protocols. Moreover, we also include very challenging AD benchmarks such as AD on the fine-grained CUB200 dataset, as well as the second industrial application of face anti-spoofing.*

2) AD Methods: AD involves classifying data into two categories: normal and anomalous. While the normal class is well-defined, the anomaly class encompasses all other variations, making it broader and more complex. This leads to a natural data imbalance. In this paper, we categorize existing methods based on the availability of anomalous samples, leading to two distinct variations: UNS-AD and IMS-AD.

There exist different UNS-AD approaches. *Pretext task methods* learn to solve a different auxiliary task on normal data [10], [11], [23], [24], and the anomaly score reflects performance on this auxiliary task. *Two-stage methods* consist of representation learning and anomaly score estimation. After training an encoder on normal data using self-supervised learning [15], [25], [26], [27] or an encoder pre-trained on additional datasets [9], [28], [29], a classifier computes the anomaly score in the latent space [1], [30], [31]. Some methods [7], [8] use a nearest neighbor queue for anomaly scoring. *Density estimation methods* estimates the normal distribution using deep density estimators like normalizing-flows [32], variational models [33] or diffusion models [34], [35]. *Reconstruction methods* measure the reconstruction error of a bottleneck encoder-decoder, trained with denoising autoencoders [36], [37] or two-way GANs [38], [39], [40]. Some incorporate memory in the latent space of an auto-encoder [5], [6] during training for optimal reconstruction. More recently, *knowledge distillation methods* have been adapted to AD by using the representation discrepancy of anomalies in the teacher-student model [41], [42].

IMS-AD mainly revolves around two-stage methods and anomaly distance methods. Some recent work has tried to generalize pretext task to UNS-AD [23]. In the *IMS-AD two-stage methods*, a supervised classifier with the anomalous samples is trained in the second stage instead of the aforementioned one-class estimator [43]. *Distance methods* directly use a distance to a centroid as the anomaly score and learn the model to maximize the anomaly distance on anomalous samples and minimize it on normal samples [44], [45]. To our best knowledge, **no** IMS-AD methods in the literature *use any kind of memory mechanism* for the anomaly score computation. For instance, MemSeg [46] and PatchCore [2] use frozen pretrained encoder and memory is not learned. Moreover, there is no anomaly distance learning.

Our method performs exceptionally well in both unsupervised (one-class) and imbalanced supervised settings, establishing it as a **unified** approach.

3) Vision-Language Models (VLM) Based AD: Recent years have seen significant advances in vision-language models (VLMs). CLIP [20] pioneered large-scale image-text pretraining, enabling strong generalization across a wide range of tasks. Models like LLaVA and GPT-4V further unify vision and language modalities to achieve human-level understanding. These VLMs have also been applied to AD.

WinCLIP [47] introduces a window-based CLIP approach with a compositional ensemble of state words and prompt templates, aiming to efficiently extract and aggregate multi-level features. KAnoCLIP [48] and AA-CLIP [49] leverage prompt learning and lightweight adapters to adapt CLIP for industrial AD without requiring anomaly samples (i.e., in UNS-AD settings). AdaptCLIP [50] further enhances robustness by learning domain-specific prompts and fine-tuning on a small set of normal samples. AnomalyGPT [51] explores the use of large-scale VLMs to tackle structural anomaly detection in industrial environments.

While effective, these methods are multimodal and rely on EK via large-scale pretrained encoders. Moreover, they have been primarily evaluated on industrial AD benchmarks, which may limit their generalizability to more diverse anomaly types. Although often categorized as UNS or zero-shot AD, from a strict perspective, VLM-based methods are not truly zero-shot, as information about anomalies may already be *implicitly* encoded in the pretrained visual or language modalities.

4) Other AD Tasks: Anomaly localization (AL) is a closely related task whose goal is to produce an anomaly heatmap. AL datasets range from defect localization [52] to surveillance

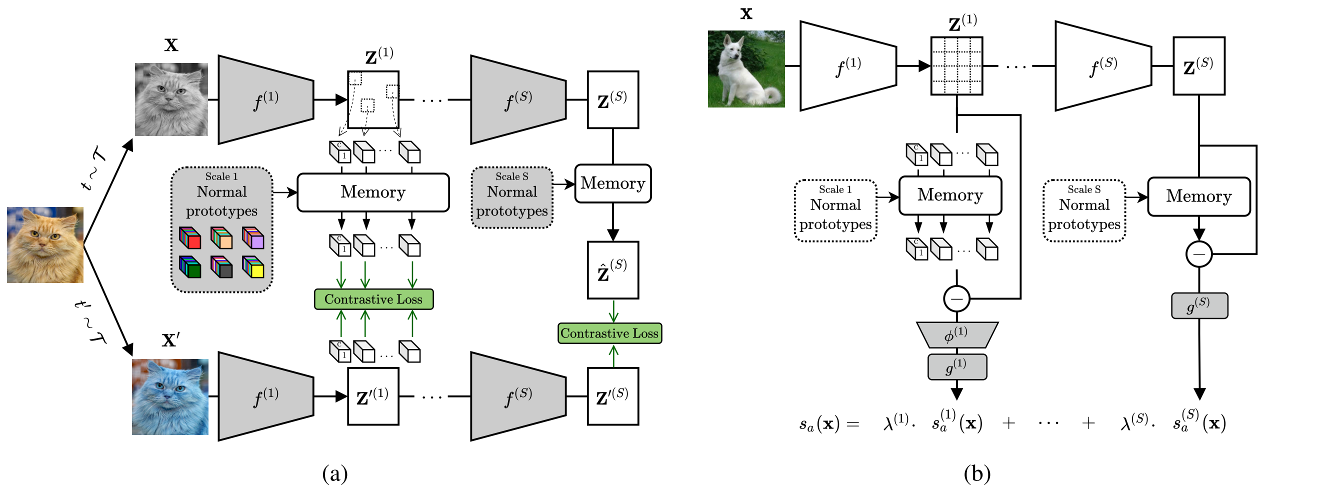

Fig. 2. Overview of AnoMem's training. (a) In the first step, representation and normal prototypes are learned simultaneously across multiple scales using Hopfield layers and contrastive learning (Sec. III-A); (b) In the second stage, multi-scale anomaly detection models are trained using memory deviation maps, a process applied ***only*** in the IMS-AD setting (Sec. III-B). Learnable components are highlighted in dark gray.

video abnormal event detection [53]. Specialized methods targeting localization [1], [54] have been introduced to efficiently solve this task.

Out-of-distribution (OOD) detection aims to detect whether or not an observation has been sampled from the same distribution as the training set [55], [56], [57]. Compared to AD, normal data of OOD is highly multi-modal since it cover multiple semantic classes. Moreover, OOD will put more weight than in AD on the acquisition process of data (e.g., capture sensor, lighting, background). Therefore, it is generally preferred for OOD detectors not to generalize too much on normal data, while it is the opposite in AD.

C. Contrastive Learning

Contrastive learning (CL) is a self-supervised representation learning method. It operates on the basis of two principles: (1) two different images should yield dissimilar representations, and (2) two different views of the same image should yield similar representations. The views of an image are characterized by a set of transformations $\mathcal{T}$. These transformations can range from geometric transformations such as crops and rotations, to colorimetric transformations with color jitter. There have been many methods enforcing these two principles: SimCLR [58], Barlow-Twins [27] and VICReg [59] with a Siamese network, MoCo [60] with a negative sample bank, BYOL [61] and SimSiam [62] with a teacher-student network or SwAV [63] with contrastive clusters. While some contrastive methods such as SimCLR and MoCo require negatives samples, other such as BYOL, SimSiam and SwAV do not.

In the simplest formulation, the pairs of representations to contrast are only considered from two views of a batch augmented by transformations (t, t'). SimCLR minimized the loss:

$$\mathcal{L}_{\text{CO}} = \frac{1}{2B} \sum_{k=1}^{B} \ell_{\text{NTX}}(z_k, {z'}_k) + \ell_{\text{NTX}}({z'}_k, z_k) \tag{2}$$

Algorithm 1 AnoMem First Learning Stage

Input: batch size B, invariance transformations $\mathcal{T}$
Initialization: encoders $f^{(1)}, \cdots, f^{(S)}$, memory $\text{HF}^{(1)}, \cdots, \text{HF}^{(S)}$.
while not reach the maximum epoch **do**
  Sample image minibatch $\boldsymbol{X}$ with labels $\boldsymbol{y}$
  Sample two invariance augmentations t, t' from $\mathcal{T}$
  Get augmented views $\mathbf{Z}^{(0)} \leftarrow t(\mathbf{X})$ and $\mathbf{Z'}^{(0)} \leftarrow t'(\mathbf{X})$
  for $s = 1 \cdots S$ **do**
    $\mathbf{Z}^{(s)} \leftarrow f^{(s)}(\mathbf{Z}^{(s-1)})$ and $\mathbf{Z'}^{(s)} \leftarrow f^{(s)}(\mathbf{Z'}^{(s-1)})$
    Sample $\lfloor H^{(s)}\ W^{(s)}\ r^{(s)} \rfloor$ vectors $Z_{i,j}^{(s)}$ from $\mathbf{Z}^{(s)}$
    Retrieve each $Z_{i,j}^{(s)}$ memory prototypes using Eq. (3) with the s^{th} scale memory layer.
  end for
  Compute $\mathcal{L}_{\text{COM-MS}}$ from Eq. (7)
  Gradient descent on $\mathcal{L}_{\text{COM-MS}}$ to update $f^{(1)}, \cdots, f^{(S)}$and $\text{HF}^{(1)}, \cdots, \text{HF}^{(S)}$.
end while
Output: Encoder network f, and the **multi-scale memory prototypes** from $\text{Mem}^{(1)}, \cdots, \text{Mem}^{(S)}$.

where z, z' are the last features of the two augmented batches and ℓ_{NTX} is the normalized temperature-scaled cross entropy loss (NTX). In practice, minimizing will yield representations with the most angular spread variance, while retaining angular invariance in regard to $\mathcal{T}$. Memory mechanisms can also be used into CL as in [64] and [65]. During training, the positive and negative pairs are augmented with the samples nearest neighbors from a memory queue. This allows the model to reach better performance for smaller batch sizes.

III. Proposed Method

Sec. III-A first details a novel training procedure to simultaneously learn an encoder's representations and a set of multi-scale normal class prototypes (the pseudocode can be found in Alogithm 1). Sec. III-B then presents how to use the encoder and the normal prototypes to train a one-class or imbalanced supervised anomaly detector in a unified framework. Our model is summarized in Fig. 2.

Notation: Let $\mathcal{X} = \{(x_k, y_k)\}_k$ be a training dataset with normal samples ($y_k = 1$) for UNS-AD, and potentially anomalies

($y_k = 0$) for IMS-AD. $f : \mathbb{R}^{H\times W\times C} \rightarrow \mathbb{R}^D$ is a backbone made of several stages $f^{(1)}, \cdots, f^{(S)}$ where the dimensions of the s^{th} scale feature map are $H^{(s)} \times W^{(s)} \times C^{(s)}$. We note $z^{(s)} = f^{(s)} \circ \cdots \circ f^{(1)}(x)$.

A. Learning Representations and Memorizing Normal Class Prototypes

We first present how memory modules can be used in CL to provide robust and representative normal class prototypes, and then generalize the idea to several scales throughout the encoder (shown in Fig. 2a). We choose to apply a CL type method rather than other unsupervised learning schemes, as it produces better representations with very few labeled data [66]. We also favor self-supervised learning only on the normal data rather than using a pre-trained encoder on generic datasets [28] which often performs poorly on data with a significant distribution shift.

1) Baseline: To learn unsupervised representations and identify a set of normal prototypes, one approach is to *sequentially* apply CL followed by *k-means clustering* and use the cluster centroids as the *normal prototypes*. However, this method has two significant limitations:

1) The steps of representation learning and prototype construction are *entirely decoupled*. Research has shown that incorporating a few representative samples in the negative examples significantly enhances representation quality and reduces the reliance on large batch sizes [63], [64].
2) The k-means prototypes often fail to capture atypical samples effectively. Consequently, harder normal samples may not be adequately represented by these prototypes, leading to a higher false rejection rate during anomaly detection (AD). *We perform ablation studies to compare AnoMem with this baseline using k-means centroids in Sec. IV-G.*

2) Integrated Method: Jointly Learning Memory and Encoder: To address above-mentioned flaws, we introduce a novel approach based on memory modules to *jointly learn* an encoder and normal prototypes. Let z_k and z'_k respectively be the encoder features for the contrastive upper and lower branch. Instead of contrasting z_k and z'_k, we apply beforehand a Hopfield memory layer HF($\cdot$) to the upper branch ***in the case of normal samples***. The memory layer is only applied when the sample is normal, since we assumed that anomalous data is significantly more variable and less defined than normal data. As such, we note

$$\text{Mem}(z, y) = y \cdot \text{HF}(z) + (1 - y) \cdot z \tag{3}$$

Thus, $\text{Mem}(z, 1) = \text{HF}(z)$ for normal samples and $\text{Mem}(z, 0) = z$ for anomalies. We introduce now the following contrasted memory loss for representation learning:

$$\mathcal{L}_{\text{COM}}(z, z', y) = \frac{1}{2B} \sum_{k=1}^{B} \Big[\ell_{\text{NTX}} \left(\text{Mem}(z_k, y_k), z'_k\right) + \ell_{\text{NTX}} \left(z'_k, \text{Mem}(z_k, y_k)\right) \Big] \tag{4}$$

where

$$\ell_{\text{NTX}}(z, z') = -\log \frac{\exp(\text{sim}(z, z')/\tau)}{\sum_{\vec{p}} \mathbb{1}_{[\vec{p} \neq z]} \exp(\text{sim}(z, \vec{p})/\tau)} \tag{5}$$

and $\vec{p}$ covers any representation inside the multi-view batch, τ is a temperature hyperparameter and $\text{sim}(\cdot, \cdot)$ is the cosine similarity.

Remarks: Unlike existing two-stage AD methods [1], [9], [15], [25], [26], [28], [29], [30], [31] which separate representation learning from anomaly score estimation, our approach ***explicitly*** incorporates anomalous and normal labels from the very beginning of the representation learning phase. ***Importantly***, the labels y_k are utilized solely to filter out anomalies from memory learning, ensuring that the representation learning is effective in both UNS and IMS settings. Additionally, the ***encoder and normal prototypes are learned simultaneously*** from the very beginning of the representation learning process.

3) Variance Loss as Regularization: Our procedure can be prone to representation collapse during the first epochs. Indeed, we observed that the dynamic between CL and the randomly initialized memory layer can occasionally lead to a collapse of all prototypes to a single point during the first epochs. To prevent this, we introduce a regularization loss to ensure the variance of the retrieved memory samples does not reach zero:

$$\mathcal{L}_{\text{V}}(z, y) = -\frac{1}{\sum_k y_k} \sum_{k=1}^{B} y_k \sqrt{\text{Var}\left[\text{Mem}(z_k, y_k)\right]} \tag{6}$$

4) Multi-Scale Contrasted Memory: To gather information from several scales, we apply our contrasted memory loss not only to the flattened 1D output z of our encoder but also to S intermediate layer 3D feature maps $z^{(1)}, \cdots, z^{(S)}$.

We add after each scale representation $z^{(s)}$ a memory layer $\text{HF}^{(s)}$ to effectively capture multi-scale normal prototypes. However, memorizing the full 3D map as a single flattened vector would not be ideal. Indeed, at lower scales we are interested in memorizing local patterns regardless of their position. Moreover, the memory would span across a space of very high dimensions. Therefore, 3D intermediate maps are viewed as a collection of $H^{(s)}W^{(s)}$ 1D feature vectors $z^{(s)}_{i,j}$ rather than a single flattened 1D vector. This is equivalent to remembering the image as patches. Since earlier features map will have a high resolution, the computational cost and memory usage of such approach can quickly explode. Thus, we only apply our contrasted memory loss on a random sample with ratio $r^{(s)}$ of the available vectors on the s^{th} scale.

Our *multi-scale contrasted memory loss:*

$$\mathcal{L}_{\text{COM-MS}} = \frac{1}{\sum_s |\Omega^{(s)}|} \sum_{s=1}^{S} \sum_{(i,j)\in\Omega^{(s)}} \lambda^{(s)} \Big[\mathcal{L}_{\text{COM}}\left(z^{(s)}_{i,j}, z'^{(s)}_{i,j}, y\right) + \lambda_V (z^{(s)}_{i,j}, y) \Big] \tag{7}$$

where λ_V controls the impact of the variance loss, $\lambda^{(s)}$ controls the importance of the s^{th} scale, and $\Omega^{(s)}$ is a random sample without replacement of $\lfloor H^{(s)}\ W^{(s)}\ r^{(s)} \rfloor$ points from $1, H^{(s)} \times 1, W^{(s)}$. We choose to put more confidence on the latest stages which are more semantically meaningful than earlier scales, meaning that $\lambda^{(1)} < \cdots < \lambda^{(S)}$.

We simultaneously minimize this loss on all the encoder stages and the S memory layers' weights. An overview of this first stage is given in Fig. 2a, and its algorithm is presented in Alg. 1. Compared to previous memory bank equipped anomaly and OOD detectors [5], [6], [7], [8], [67], our model is the first to memorize the normal class at several scales, allowing it to be more robust to anomaly sizes. Moreover, the use of normal memory does not only improve AD but also the quality of the learned representations, as will be discussed in Sec. IV-G.

B. Inference: Multi-Scale Normal Prototype Deviation

In this second step, our goal is to compute an anomaly score given the pre-trained encoder f and the multi-scale normal memory layers $\mathrm{HF}^{(1)}, \cdots, \mathrm{HF}^{(S)}$. For each scale s, we consider the difference $\Delta^{(s)}$ between the encoder feature map $z^{(s)}$ and its recollection from the s^{th} memory layer. The recollection process consists in spatially applying the memory layer to each $C^{(s)}$ depth 1D vector:

$$\Delta^{(s)} = z^{(s)} - \mathrm{HF}^{(s)}\left(z^{(s)}\right) \tag{8}$$

where $(\mathrm{HF}(z))_{i,j} = \mathrm{HF}(z_{i,j})$.

Remark: It is worth noting that ***only this second stage has to be swapped*** between UNS and IMS settings, resulting in a unified easily-switchable framework for AD.

a. Unsupervised AD. We use the L_2 norm of the difference map as an anomaly score for each scale, and ***no further training*** is required:

$$s_a{}^{(s)}(x) = \|\Delta^{(s)}\|_2 \tag{9}$$

b. Imbalanced supervised AD. The second training stage of IMS-AD is summarized in Fig. 2b. We use the additional anomaly data to train S scale-specific classifiers on the difference map $\Delta^{(s)}$. Each classifier is first composed of an average pooling layer ϕ followed by a two-layer MLP $g^{(s)}$ with a single scalar output. ϕ reduces the spatial resolution of $\Delta^{(s)}$, to prevent using very large layers on earlier scales. The output of $g^{(s)}$ directly corresponds to the s^{th} scale anomaly distance:

$$s_a{}^{(s)}(x) = g^{(s)} \circ \phi(\Delta^{(s)}) \tag{10}$$

Each scale-specific classifier is trained using the intermediate features of the same normal and anomalous samples used during the first step along their labels. The training procedure is similar to other distance-based anomaly detectors [44], [45] where the objective is to obtain small distances for normal samples while keeping high distances on anomalies. We note that our model is the *first to introduce memory prototypes learned during representation learning into the anomaly distance learning*. The distance constraint is enforced via a double-hinge distance loss:

$$\ell_{\text{dist}}(d, y) = y\cdot \max\left(d - \frac{1}{M}, 0\right) + (1-y)\cdot \max(M - d, 0)$$

where d is the anomaly distance for a given sample and M controls the size of the margin around the unit ball frontier. Using this loss, both normal samples and anomalies will be correctly separated without encouraging anomalous features to be pushed toward infinity. Our ***second stage loss*** is:

$$\mathcal{L}_{\text{SUP}} = \frac{1}{S \cdot B} \sum_{k=1}^{B} \sum_{s=1}^{S} \ell_{\text{dist}}\left(g^{(s)} \circ \phi(\Delta^{(s)}), y_k\right) \tag{11}$$

Finally, all scale anomaly scores are merged using a sum weighted by the confidence parameters $\lambda^{(s)}$:

$$s_a(x) = \frac{1}{\sum_s \lambda^{(s)}} \sum_s \lambda^{(s)} \cdot s_a{}^{(s)}(x) \tag{12}$$

The anomaly score s_a effectively combines the expertise of the scores from each scale, making it more robust to different sizes of anomalies than other detectors. As mentioned in Sec. III-A, the $\lambda^{(s)}$ will put more weight to later scales, making our detector more sensitive to broad object anomalies.

IV. Experiments

We evaluate the performance of AnoMem across a wide range of benchmarks, demonstrating that it is the first unified approach capable of effectively addressing both UNS and IMS anomaly detection, as well as various types of anomalies, including object, local, fine-grained anomalies, and industrial applications. In Secs. IV-A and IV-B, we provide a detailed overview of the benchmarks and implementation details. Subsequently, Secs. IV-C to IV-F compare our method with the state-of-the-art (SoTA) under different scenarios. Finally, Secs. IV-G to IV-K present various ablation studies.

Remarks: Since not all existing methods were fully evaluated in all settings and for a fair comparison under the same conditions, we either use existing implementations or re-implement and evaluate them ourselves for many of those methods. We denote those results in blue in the following tables. As posited in [4], it is ***unfair*** *to directly compare methods trained from scratch, including AnoMem, with those utilizing external knowledge (EK)*, such as pre-trained models or additional datasets. Thus, we distinguish between non-EK and EK methods, exemplified in Tabs. II to IV.

A. Evaluation Protocol and Benchmarks

We evaluated AnoMem across a broad range of anomaly detection benchmarks, as outlined in Tab. I. This evaluation encompassed: (1) one-class novelty detection, adopting the terminology from [41], with both object-level and fine-grained anomaly detection, (2) two real-world applications of industrial inspection and face anti-spoofing (FAS), and (3) out-of-distribution (OOD) detection. In total, we examined seven distinct datasets.

1) **One-class novelty detection:** one class is designated as normal, while all other classes are treated as anomalous or novel. The final result reported is the mean of all runs for each possible normal class. We varied the ratio γ of anomaly data in the training set and averaged the metrics over 10 random samples for each ratio. The UNS-AD setting, where $\gamma = 0$, is a special case of the IMS-AD setting. This protocol presents a *realistic challenge* and is commonly used in anomaly detection

TABLE II

COMPARISON WITH SOTA FOR ONE-CLASS NOVELTY DETECTION. THE UNS-AD SETTING, WHERE $\gamma = 0$, IS A SPECIAL CASE OF THE IMS-AD SETTING. * IS FROM [35]. NOTE THAT, EK METHODS IN THE UNS SETTING USE VERY LARGE EXTERNAL DATASETS, SIGNIFICANTLY MORE THAN OUR IMS SETTING WHERE WE ACHIEVED BETTER RESULTS (SEE TEXT)

		AUROC (%)	CUB-200				CIFAR-10				CIFAR-100			
		Models \ γ	0.	0.01	0.05	0.10	0.	0.01	0.05	0.10	0.	0.01	0.05	0.10
without EK	UNS	MemAE [5]	59.6				60.9				57.4			
		PIAD [38]	63.5				79.9				78.8			
		GOAD [11]	66.6				88.2				74.5			
		MHRot [10]	77.6				89.5				83.6			
	IMS	Supervised		53.1	58.6	62.4		55.6	63.5	67.7		53.8	58.4	62.5
		Elsa [43]		77.8	81.3	82.9		80.0	85.7	87.1		81.3	84.6	86.0
	Unified	DeepSAD [44]	53.9	62.7	63.4	65.1	60.9	72.6	77.9	79.8	56.3	67.7	71.6	73.2
		DP VAE [33]	61.7	65.4	67.2	69.6	52.7	74.5	79.1	81.1	56.7	68.5	73.4	75.8
		AnoMem (ours)	**81.4**	**84.1**	**85.3**	**86.0**	**91.5**	**92.5**	**97.1**	**97.6**	**86.1**	**90.9**	**92.3**	**94.7**
with EK		RD [41]	-				86.5				-			
		SSD [30]	-				90.0				85.1			
		CSI [15]	52.4*				94.3				85.8			
		MSC [9]	-				94.8				94.4			

TABLE III

PERFORMANCE ON INDUSTRIAL DEFECT DETECTION USING THE MVTEC-AD DATASET

(a) **Without EK**, such as pre-trained models or extra datasets.

Model	MHRot	CutPaste [1] CVPR'21	PatchCore [2] CVPR'22	NSA [3] ECCV'22	SPR [4] ICCV'23	AnoMem (UNS)
AUC	86.3	96.0	92.1	97.2	97.7	97.9

(**b**) **With EK**. Note that, although IMS-AnoMem is included in this table, ***it does not rely on additional large-scale datasets***. Our IMS-AnoMem uses ***significantly less*** data than other EK methods.

Model	DRAEM [12] ICCV'21	RD [41] CVPR'22	PatchCore CVPR'22	UniAD [22] NeuRIPS'22	MSC [9] AAAI'23	DeSTS [14] CVPR'23	FastViT [76] + [13] ICCV'23 + PAMI'24	AnoMem (IMS) (***"not really EK"***) (γ=.01)	(γ=.05)	(γ=0.1)
AUC	98.0	98.5	99.1	98.6	87.2	97.0	97.3	98.4	98.7	98.9

TABLE IV

PERFORMANCE ON FACE PRESENTATION ATTACK DETECTION USING THE WMCA DATASET, COLUMNS REPRESENT UNSEEN ATTACKS

		AUROC (%)	WMCA				
		Models / Kind	All	PP	SR	PM	FM
without EK	UNS	PIAD [38]	76.4				
		MHRot [10]	81.3				
		GOAD [11]	86.1				
	IMS	Supervised		78.3	77.1	80.7	81.9
		Elsa [43]		86.1	84.3	89.2	89.1
		SadCLR [45]		89.8	88.5	92.7	91.9
	Unified	DP VAE [33]	53.9	-	-	-	-
		DeepSAD [44]	71.2	79.9	80.3	81.8	83.4
		AnoMem (ours)	**86.9**	**91.3**	**89.8**	**93.0**	**92.7**
w/ EK		CSI [15]	82.7				
		MSC [9]	85.3				

literature to assess the model's ability to *generalize* to various anomalies *without* relying on a *significant amount* of training anomalies. Datasets included CIFAR-10, CIFAR-100 [68], and CUB-200 [69]. Notably, CUB-200, with its 200 fine-grained bird classes (each with about 50 images), presents a particularly challenging scenario due to its subtle style and local anomalies.

2) **Real-world applications:** we focus on two real-world applications of AD, i.e., industrial inspection detection (IDD) using the MVTec-AD dataset [52] and face anti-spoofing (FAS) using the WMCA dataset [70]. The MVTec dataset includes 5 texture classes and 10 object classes. The training set comprises 3,629 images free of anomalies, while the test set encompasses 1,725 images, containing both anomalous and anomaly-free instances. Each class includes various defects to enable comprehensive testing. AL evaluation of this dataset is beyond the scope of this paper and will be addressed in future work. Shifting to the FAS evaluation protocol, which aims to differentiate real from fake faces, the training and test data come from the same dataset, but one attack type is left unseen during training. This setup allows us to evaluate the model's generalization and robustness against new attack types. The WMCA dataset, containing over 1900 RGB videos, includes various attacks like paper print (PP), screen recording (SR), paper mask (PM), and flexible mask (FM), covering object, style, and local anomalies.

3) Out-of-distribution (OOD) detection: we assessed how well the model can distinguish between in-distribution (ID) and out-of-distribution (OOD) samples after training on the ID dataset. For this evaluation, we included the commonly used SVHN [71] and LSUN [72] datasets.

We report the area under the ROC curve (AUROC) averaged over all normal classes.

B. Implementation Details

1) Optimization: Training is performed under SGD optimizer with Nesterov momentum [73], using a batch size of $B = 1024$ and a cosine annealing learning rate scheduler [74] for both of the stages.

2) Model Design: Regarding network architecture, a Resnet-50 [75] ($\approx 25M$ parameters) is used as the backbone f. We consider two different memory scales: one after the third stage and another after the last stage. The associated memory layers are respectively of size $N_{\text{Mem}}^{(1)} = 512$ and $N_{\text{Mem}}^{(2)} = 256$ with an inverse temperature $\beta = 2$, along with a pattern sampling ratio of $r^{(1)} = 0.3$ and $r^{(2)} = 1$. The choices of memory size and sampling ratios are respectively discussed in Sec. IV-H and Sec. IV-I. The scale confidence factors are set to increase exponentially as $\lambda^{(s)} = 2^{s-1}$, and the variance loss factor is fixed to $\lambda_V = 0.05$ after optimization on CIFAR-10. We use $\tau = 0.1$ as suggested in [58]. For the anomaly distance loss, we choose a margin size of $M = 2$. As for the inverse temperature β, we used the value recommended in [17] based on the size of our Hopfield memory size.

We conducted a performance evaluation of AnoMem using different backbone architectures, including EffNet-B0, ResNet18, and ResNet50. In the context of one-vs-all evaluation settings (UNS and IMS) on the CIFAR dataset, we achieved consistently high performance.

Remarks: EffNet-B0 emerged as the top performer, demonstrating superior results, followed closely by ResNet50 and then ResNet18. The performance gaps between these architectures were relatively small, with EffNet-B0 outperforming ResNet50 by a margin of 0.2% to 0.3%, while ResNet50 surpassed ResNet18 by a modest margin of 0.4% to 0.6%. In the rest of the paper, we report the performance of AnoMem using ResNet50, which was the median performer. It is worth noting that the memory modules are scalable with the chosen backbone.

3) Input Preprocessing: For all datasets except CIFAR-10 and CIFAR-100, we resize the input images to 224×224. Prior works on the MVTec-AD often resized the images to 128×128 or 256×256, and [41] found that the resolution of 256×256 produced better results (up to 2 points). We observed the same findings in our experiments; the resolution of 256×256 gives slightly better results than 224×224. However, to demonstrate our method as a generic detector, we resize all the images to the same size of 224×224. As for the contrastive invariance transformations, we use random crop with rescale, horizontal symmetry, brightness jittering, contrast jittering, saturation jittering with Gaussian blur and noise as in SimCLR [58].

Remarks: It is worth noting that we do not need specific augmentations as in CSI [15]. Indeed, CSI requires *additional* shift transformations (alongside standard augmentations like SimCLR and AnoMem) to generate its pseudo OOD negatives. AnoMem performs *differently*: it directly learns prototypes of normal samples and does *not* rely on pseudo negatives or require shift augmentations.

C. Results on Novelty Detection

For non-EK methods, we consider various categories of unsupervised (UNS) approaches: the reconstruction error generative model [38], the knowledge distillation method [41], methods based on pretext tasks [10], [11], and the two-stage method [15]. Additionally, we include the two-stage IMS-AD method in [43]. To further illustrate the limitations of classical binary classification, we include a traditional deep classifier trained with balanced batches of normal and anomalous samples. We also incorporate unified methods applicable in both unsupervised and supervised settings, such as a reconstruction error model [33], and direct anomaly distance models [44]. Finally, we consider EK methods, including RD, SSD, CSI, and MSC. The results are summarized in Table II, from which we can draw several key observations:

1) **Methods without EK:**
   - ***Superiority of AnoMem***: First, among the methods without EK, the classical supervised approach significantly underperforms compared to specialized anomaly detectors across all datasets. This underscores the necessity of using dedicated AD methods, as classical models tend to overfit on anomalies. Furthermore, AnoMem consistently outperforms all other detectors considered in the study. While its performance improves with the availability of more abnormal data, it remains highly competitive even when trained with only normal samples.
   - ***Efficient memory usage in AnoMem***: the memory usage in AnoMem is significantly more efficient than that in MemAE or similar models [5], [6], which rely on pixel-wise reconstruction loss to learn memory. In contrast, AnoMem employs contrastive learning, allowing its learnable neural network-based memory to develop more robust representations. Unlike other methods, such as coreset technique, which are limited by static memory mechanisms, AnoMem creates normal prototypes that are less constrained and more semantically rich and generalizable. This approach enables AnoMem to achieve better performance and adaptability in AD tasks.
2) **Comparison with EK methods:**
   - ***Remarks.*** Although EK methods such as CSI and MSC do not directly use anomalies, they leverage extensive additional datasets (like ImageNet1K or 21K with up to 14 million images). The pre-trained models in these methods incorporate significant information about "pseudo" anomalies (images from the additional dataset), while the "strict" UNS setting operates under the assumption of no prior knowledge.

- ***Efficiency of AnoMem.*** Despite requiring less training data, AnoMem demonstrates superior efficiency. For instance, on CIFAR-10, MSC achieved 94.8% AUC with the help of 1 million additional images, while IMS-AnoMem attained 97.1% AUC using only 5% of anomaly classes. This highlights AnoMem's effectiveness and efficiency in AD even with minimal data.
- ***Generalization capacity on challenging benchmark datasets.*** AnoMem significantly outperforms CSI on CUB-200 and MSC on FAS and industrial inspection tasks, achieving up to 10 percentage points higher AUC (details provided in the next section). This indicates that SoTA pre-trained methods like CSI and MSC may perform poorly in the presence of distribution shifts, whereas AnoMem consistently performs excellently. Notably, as other detectors, including the ***SoTA CSI, struggle with CUB-200***, this dataset represents a valuable benchmark for assessing AD performance.

3) **Unified anomaly detector:** AnoMem performs very well in both UNS-AD and IMS-AD while other unified methods fail in UNS-AD. AnoMem is the ***first efficient unified anomaly detector***. We also note that our change from UNS-AD to IMS-AD was done with minimal hyperparameter tuning. This is due to the first training step being shared between two settings.

D. Results on Industrial Defect Detection

Similar to [4] and the previous section, Table III is divided into two distinct parts to address the discrepancy of non-EK and EK methods. From this table, several key observations can be made:

1) Performance of UNS-AnoMem. ***Without*** any specific design for industrial inspection, UNS-AnoMem outperforms all previous methods that do not use external knowledge (EK), including PatchCore [2], NSA [3], and SPR (score-based perturbation resilience) [4].
2) IMS-AnoMem performance in comparison with methods using EK.
   - PatchCore [2] and RD [41] utilize pre-trained models, whereas DRAEM [12] generates synthetic anomalies from an additional dataset instead of relying on pre-trained models. Analyzing the outcomes of PatchCore, which has two variations, with and without EK, indicates that incorporating EK enhances performance. Trained from scratch with a limited amount of anomalies, IMS-AnoMem achieves SoTA performance. The three results shown correspond to different ratios of anomalous samples used ($\gamma = 0.01, 0.05, 0.10$).
   - Other detectors are more complex than ours. DeSTS [14] combines a pre-trained teacher network, a denoising student encoder-decoder, and a segmentation network into a framework, while UniAD [22] utilizes a pre-trained Transformer. "FastViT + [13]" combines three methods, that are FastVit in [76], NSA in [3], and SSMCTB (self-supervised masked convolutional transformer block) in [13], but still perform worse than AnoMem, even in the UNS setting.

TABLE V
PERFORMANCE FOR OOD DETECTION

AUROC (%)	CIFAR-10 (ID) →		
Models / OOD data	SVHN	LSUN	CIFAR-100
GOAD [11]	96.3	89.3	77.2
MHRot [10]	97.8	92.8	82.3
FS [77]	95.3	96.2	84.8
Energy OE [78]	96.6	94.0	84.4
Energy PEBAL [79]	97.7	94.5	85.2
CSI [15]	**99.8**	97.5	89.2
AnoMem (ours)	99.2	**97.5**	**89.2**

E. Results on Face Anti-Spoofing (FAS)

Table IV compares our model's performance on the FAS intra-dataset cross-type protocol[2] with the methods discussed in Sec. IV-C. ***Without*** any additional tuning specific to face data, our method achieves high-quality results on the FAS task, surpassing existing anomaly detection models across all unseen attack types. Notably, it reduces the performance gap between coarse attacks (e.g., PM, FM) and more challenging fine-grained attacks (e.g., PP, SR), thanks to its multi-scale anomaly detection approach. AnoMem consistently outperforms both CSI and the pre-trained MSC model. Additionally, the use of certain large-scale datasets, such as ImageNet-21K, may not be feasible in industrial settings due to restrictive licensing terms.

F. Results on Out-of-Distribution (OOD) Detection

As shown in Tab. V, AnoMem performs comparably to the SoTA baseline CSI [15]. However, CSI relies heavily on storing the entire training set features during inference, resulting in a substantial memory footprint with large training sets. In contrast, our model efficiently learns a *fixed* size memory.[3] Additionally, CSI tends to perform poorly on datasets with a limited number of training normal samples, as evidenced by the results on the one-vs-all CUB-200 (with 50 images per class).

It is important to note that the algorithms presented in [80] achieve higher performance on this task, but they rely on models pre-trained on a large-scale dataset such as ImageNet-21K (***14M*** images) [81]. Therefore, a direct comparison between these methods and AnoMem, which is trained from scratch on a smaller dataset (***50K*** images from CIFAR),

[2]Only a few studies have explored applying **generic** anomaly detection to this complex real-world problem involving local anomalies. Consequently, we need to replicate all the results shown in Tab. IV.

[3]In terms of memory usage, CSI and pre-trained models using KNN require storing the entire normal dataset, whereas AnoMem only needs a *fixed* size. For example, for OOD detection with CIFAR as the ID dataset, CSI must retain 60000 samples. In contrast, AnoMem requires only 768 samples (256 for one memory module and 512 for another).

TABLE VI
PERFORMANCE OF ANOMEM FOR IMAGE CLASSIFICATION THROUGH LINEAR EVALUATION

Memory	Multi-scale	CIFAR10	CIFAR100
-	-	88.2	80.5
✓	-	91.4 (+3.2)	84.1 (+3.6)
✓	✓	**91.9** (+0.5)	**84.8** (+0.7)

TABLE VII
PERFORMANCE OF ANOMEM FOR UNS-AD WITH MULTI-SCALE MEMORY

Memory	Multiscale	CIFAR10	CIFAR100	WMCA
-	-	88.1	82.7	81.6
-	✓	89.4 (+1.3)	83.8 (+1.1)	84.0 (+2.4)
✓	-	90.5 (+2.4)	85.3 (+2.6)	84.8 (+3.2)
✓	✓	**91.5** (+1.0)	**86.1** (+0.8)	**86.9** (+2.1)

is not entirely fair. The pre-training samples used in [80] overlap in nature with the anomalous samples that need to be detected. Additionally, this protocol involves a normal class composed of several subclasses, demonstrating the capability of our memory modules to handle multi-modal normal cues effectively.

G. Impact of the Multi-Scale Memory

This section examines the impact of multi-scale memory layers and highlights their benefits for our model's performance. We will illustrate these advantages through two distinct tasks: image multi-class image classification and anomaly detection. The baseline is the model using only the last feature map, with k-means centroids as its normal prototypes, as detailed in Sec. III-A.

1) Image Classification: We evaluate the impact of memory on image classification using linear evaluation on standard CIFAR-10/100 benchmarks. This involves training an additional linear classifier on frozen encoder representations to assess how memory influences the contrastive learning of these representations. Unlike AD, this experiment focuses on multi-class image classification and aims to validate the effectiveness of the first step in our approach, as depicted in Fig. 2a. Tab. VI shows that incorporating memory layers in the first branch significantly enhances the quality of the encoder representations. We hypothesize that, similar to the findings in [64], adding prototypical samples in one branch enables the model to contrast positive images against representative negatives. This approach reduces the need for large batch sizes and significantly lowers multi-scale CL memory usage.

2) Anomaly Detection: To support the importance of memory for AD, we compare the anomaly detector with k-means centroids or with the normal prototypes learned during the first stage (Table VII). In the first case, we train the anomaly detectors with the same procedure but instead of fetching the Hopfield layer output we use the closest k-means centroid. Lastly, we measure the impact of multi-scale AD by comparing the single-scale model only using the last feature map, and our

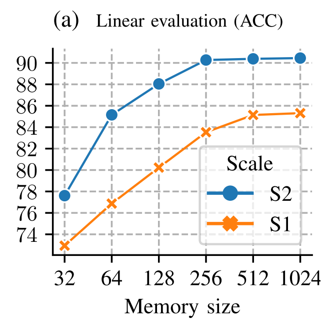

Fig. 3. Memory experiments on CIFAR10.

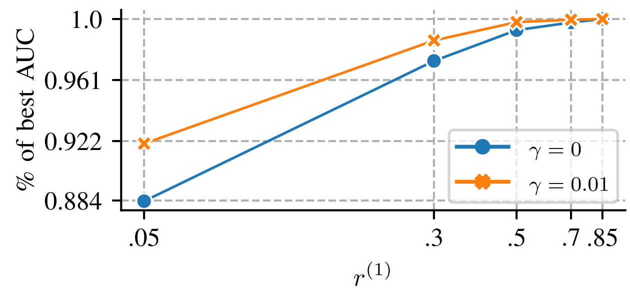

Fig. 4. Sampling ratio experiments.

two-scale model. While we sacrifice some of the memory and training time for the additional scale, our AD performance is improved significantly.

H. Impact of Memory Size and Scale

The memory size must be carefully chosen at every scale to reach a good balance between the normal class prototype coverage and the memory usage during training and inference. This section presents some rules of thumb regarding the sizing of memory layers depending on the scale.

We start by plotting the relation between the last scale memory size, the representation separability, and the final AD performance in Fig. 3a. As one could expect, higher memory size produces better quality representations and more accurate anomaly detector. However, we can note that increasing the memory size above 256 has significantly less impact on the anomaly detector performance. Therefore a good trade-off between memory usage and performance seems to be at 256 on CIFAR-10.

Further, we study the impact of feature map scale on the required memory size. In Fig. 3b we fix the last scale memory size to 256 and look at the AD accuracy for various ratios of memory size between each scale. As lower scales benefit more from a larger memory than higher scales, we hypothesize that the feature vectors of more local texture-oriented features are richer and more variable than global object-oriented ones. Memory layers thus need to be of higher capacity to capture the complexity.

In Tab. VIII, we compare the memory usage of different memory module types across two datasets. Notably, pretrained models using a nearest-neighbor (NN) queue require storing the entire normal set, while AnoMem maintains a fixed memory size.

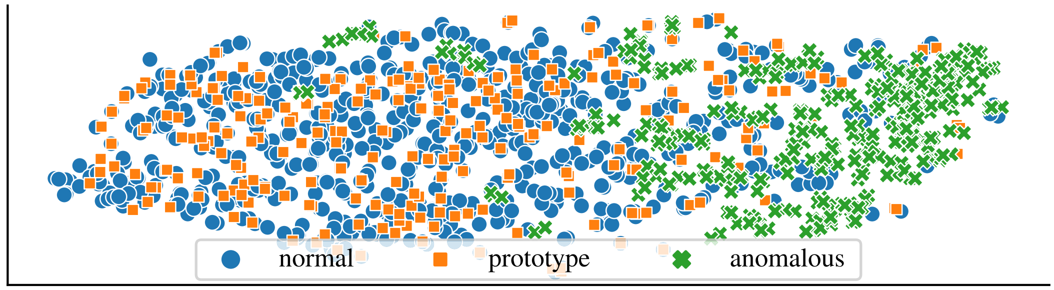

Fig. 5. t-SNE of learned prototypes, normal and anomalous samples on CIFAR10.

TABLE VIII
COMPARISON OF ADDITIONAL MEMORY USAGE FOR STORING NORMAL PROTOTYPES

Model	CSI	MSC	MemAE	AnoMem S1 (ours)	AnoMem S2 (ours)
Mem Type	NN queue	NN queue	AE	Hopfield	Hopfield
CIFAR-10	117.2MB	468.8MB	1.2MB	2.0MB	4.0MB
WMCA	227.5MB	909.8MB	1.2MB	2.0MB	4.0MB

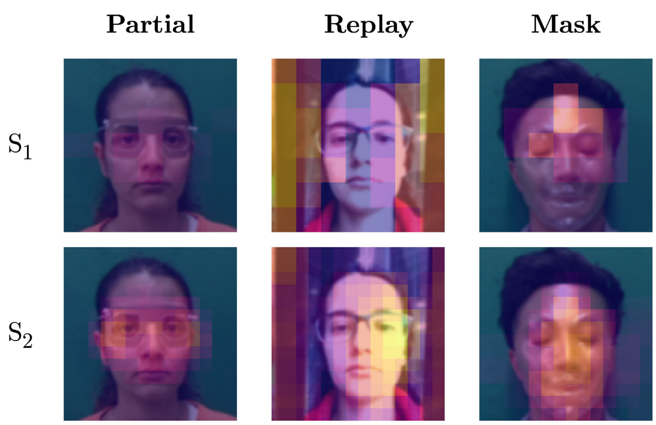

Fig. 6. Visualization of rescaled normal prototype deviation maps at scale S_1 (7×7) and S_2 (14×14) across different spoof types in the WMCA dataset.

We also compare the effect of the scale on the performance of our model. As shown in Fig. 3b, the AD performance consistently improves with higher scales, even when the memory size remains the same. The AUC can increase by up to 6 points when the model is extended to scale $S2$. We also note that, even with a larger memory, performance at the lower scale $S1$ plateaus significantly below that of $S2$. This would support that higher scale features are necessary to better discriminate normal samples from anomalies.

Remarks: We believe that the optimal choice of S depends on the type of anomaly being considered. If the focus is solely on object-level anomalies, where the semantic class differs (e.g., a dog versus a cat), a smaller S is sufficient to capture the normal semantic patterns at higher feature levels. However, if finer-grained style anomalies are also taken into account, where the object belongs to the same semantic class but exhibits stylistic differences (e.g., a painting of a dog versus a real dog), a larger S is beneficial. In this case, combining high-level semantic features with low-level textural patterns becomes important, albeit at the cost of increased computational complexity. This complementarity across scales is illustrated in Fig. 6, where each scale appears to specialize in detecting different spoof-related cues.

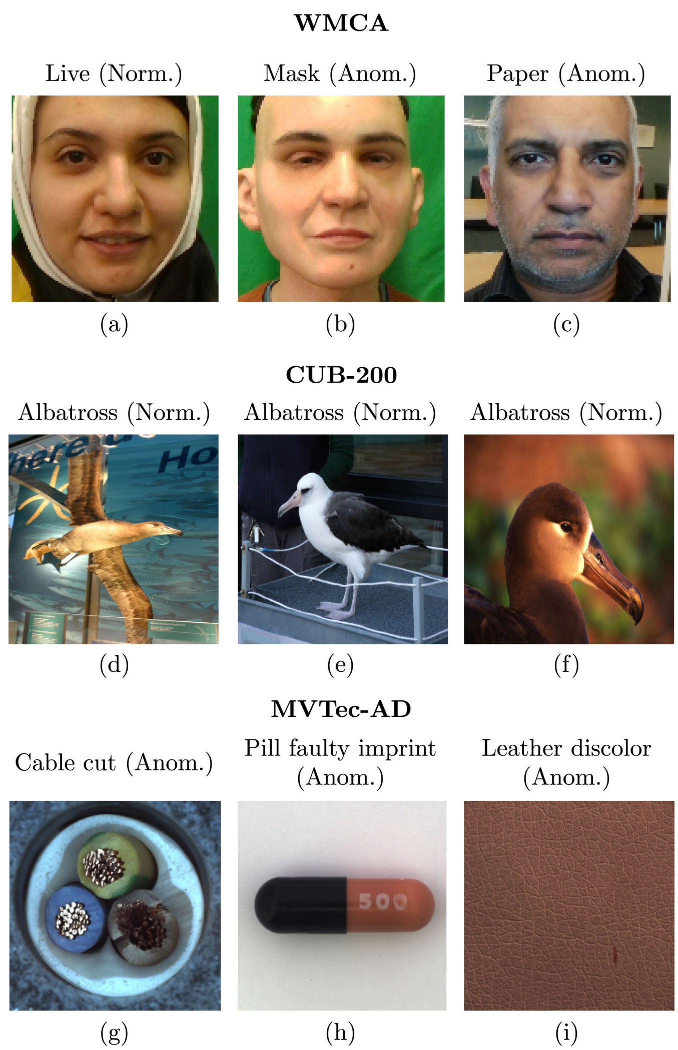

Fig. 7. Failure case visualizations across datasets. Parentheses indicate the misclassified ground truth ("Norm" = normal misclassified as anomaly, and vice versa). Cases illustrate limitations such as background sensitivity, scale variance, and subtle anomalies.

TABLE IX
LINEAR EVALUATION ON THE CIFAR-10 IMAGE CLASSIFICATION BENCHMARK FOR DIFFERENT λ_V

λ_V	0.1	0.05	0.025	0.01
AUROC (%)	91.90	91.91	91.87	91.86

TABLE X
UNS-AD EVALUATION ON CIFAR-10 AND CIFAR-100 FOR DIFFERENT $\lambda^{(s)}$ FUNCTIONS

$\lambda^{(s)}$	**CIFAR-10**	**CIFAR-100**	**avg.**
linear	91.2	85.6	88.4
exp	91.5	86.1	88.8

I. Impact of Spatial Sampling Ratios

In the first step, sampling ratios r are introduced to reduce the number of patterns used in the contrastive loss, which in turn decreases the size of the similarity matrix. At lower scales, neighboring samples are sufficiently similar, making it less detrimental to omit some patterns during training. To determine the optimal sampling ratio, we evaluate the anomaly detection AUC across various sampling ratios and anomalous data ratios γ. Given that the feature maps at the final scale are spatially small, we only adjust the sampling ratio at the first scale $r^{(1)}$ while keeping $r^{(2)} = 1$. The batch size remains fixed throughout all experiments. The results, shown in Sec. IV-I, reveal that low sampling ratios ($\gamma < 0.3$) substantially degrade AD performance. However, increasing the sampling ratio beyond a certain point offers diminishing returns: doubling the number of sampled patterns yields only a 2% improvement in relative AUC, which may not justify the added computational cost.

J. Selection and Tuning of Hyperparameters

The variance loss weight λ_V was evaluated on the CIFAR10 image classification benchmark using a range of commonly used. As shown in Tab. IX, AnoMem exhibits minimal sensitivity to λ_V. Moreover, in our tests, it remains *fixed* for all other datasets.

For the confidence weights $\lambda^{(s)}$, we compared two monotonically increasing functions linear and exponential. As we can see from the results on UNS-AD in Tab. X, the exponential weighting yields a better anomaly detection than the linear weights. We believe that the linear relation does not sufficiently enhance object anomalies and imposes excessive constraints during representation learning, particularly when contrasting very low-scale patterns.

The margin M enforces a norm constraint on normal and anomaly representations within a radius of less than $1/M$ and more than M, respectively. A value of 1 pushes all representations indiscriminately onto the unit sphere, while a high value leads to an ill-posed distance loss. We chose $M = 2$ as a compromise, allowing half of the unit sphere to be allocated for normal samples. Indeed, we are currently working on dynamically adjusting the value of M.

K. Qualitative Effectiveness of Learned Prototype

We perform in Fig. 5 a t-SNE analysis on the last-scale prototypes, along with test samples from normal and anomalous classes of CIFAR-10 after the first learning stage of AnoMem. As we can see, the representations of normal and anomalous samples are well separated, confirming the effectiveness of our memory backed representation learning. Moreover, the learned normal prototypes are well representative of the normal class, as testified by the close overlap in representation space.

L. Additional Visualizations

In Fig. 6, we additionally show the normal prototype deviation maps at two scales, S_1 and S_2, overlaid on various spoof images from WMCA. While the deviation maps are generally spatially aligned with the anomalous regions, the significant differences in score ranges and focus across scales make it non-trivial to fuse them into a single, meaningful heatmap.

- **Partial spoof:** undetected at the higher scale S_1, but fine textural inconsistencies are captured at S_2.
- **Replay spoof:** S_1 detects screen borders, while S_2 captures unnatural skin saturation.
- **Mask spoof:** S_1 identifies strong reflections, whereas S_2 reveals subtle mask texture.

The combination of both scales enhances the model's versatility in detecting diverse spoofing artifacts.

We further present several failure cases for each dataset in Fig. 7, highlighting a few notable limitations:

- **Background sensitivity:** The anomaly score can be overly influenced by the background, as seen in CUB bird images with unusual backgrounds (d, e) or WMCA Live samples with atypical clothing (a).
- **Lack of scale invariance:** The model's reliance on low-scale features can hinder its ability to generalize across different subject-to-camera distances (e.g., close-up bird image in CUB (f)).
- **Subtle anomalies:** Detection can fail when the anomalous region is too small relative to the minimum scale considered (e.g., WMCA Paper with a thin border (c), or minor defects in MVTec samples (g, h, i)).

V. CONCLUSION AND FUTURE DIRECTIONS

In this paper, we introduce a novel two-stage AD model, AnoMem, designed to seamlessly operate in both UNS and IMS settings. Our approach involves the memorization of prototypes for normal class instances to compute an anomaly deviation score. By incorporating learnable memory layers for normal instances within a CL framework, we jointly enhance encoder representations and establish robust memorization of normal samples. Subsequently, these learned normal prototypes are leveraged to train a straightforward detector within a unified framework that accommodates both UNS-AD and IMS-AD. Moreover, we extend these prototypes to multiple scales, enhancing the model's robustness against various anomaly sizes. Finally, we evaluate the performance of AnoMem on diverse datasets containing object, style, and local anomalies to demonstrate its efficiency, even when trained from scratch under limited data regimes.

Additionally, we could explore using our model for anomaly localization. This includes investigating more efficient methods for integrating multiple scales and merging the resulting anomaly maps into a unified heatmap. We might also examine the use of memory networks for non-contrastive learning approaches, such as substituting the ℓ_{NTX} loss with cross-correlation and considering techniques like Barlow-Twins as an alternative to SimCLR.

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